CYBERSTER INTERNSHIP REPORT

WEEK 1 TASKS REPORT

CYBERSTER INTERNSHIP REPORT

ROLL NO:CSI-B1-303

ARSALAN

DAY:1

Internship Report: Day 1 – Networking Fundamentals

1. Objective

The objective of Day 1 was to master the fundamentals of network behavior and protocol analysis. This ensures a solid foundation for analyzing security alerts in the coming weeks, as a large portion of SOC analysis relies on understanding underlying network protocols.

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2. Network Protocol Summary

Instead of just memorizing layers, I focused on how data travels between machines. Below is a summary of the core protocols I reviewed:

- **TCP/IP & OSI Models: I traced web requests to understand how data is encapsulated at each stage of travel.**
- **Subnetting & Ports: I practiced determining network boundaries using CIDR notation and reviewed essential SOC ports:**
  - **22 (SSH), 53 (DNS), 80/443 (HTTP/HTTPS), 445 (SMB), and 3389 (RDP).**
- **Core Protocols Analyzed:**
  - **ARP: Resolves IP addresses to MAC addresses.**
  - **DNS: The "phonebook" of the internet, converting names to IPs.**
  - **HTTP/HTTPS: The foundation of web data exchange.**
  - **ICMP: Used for diagnostic tools like 'ping'.**

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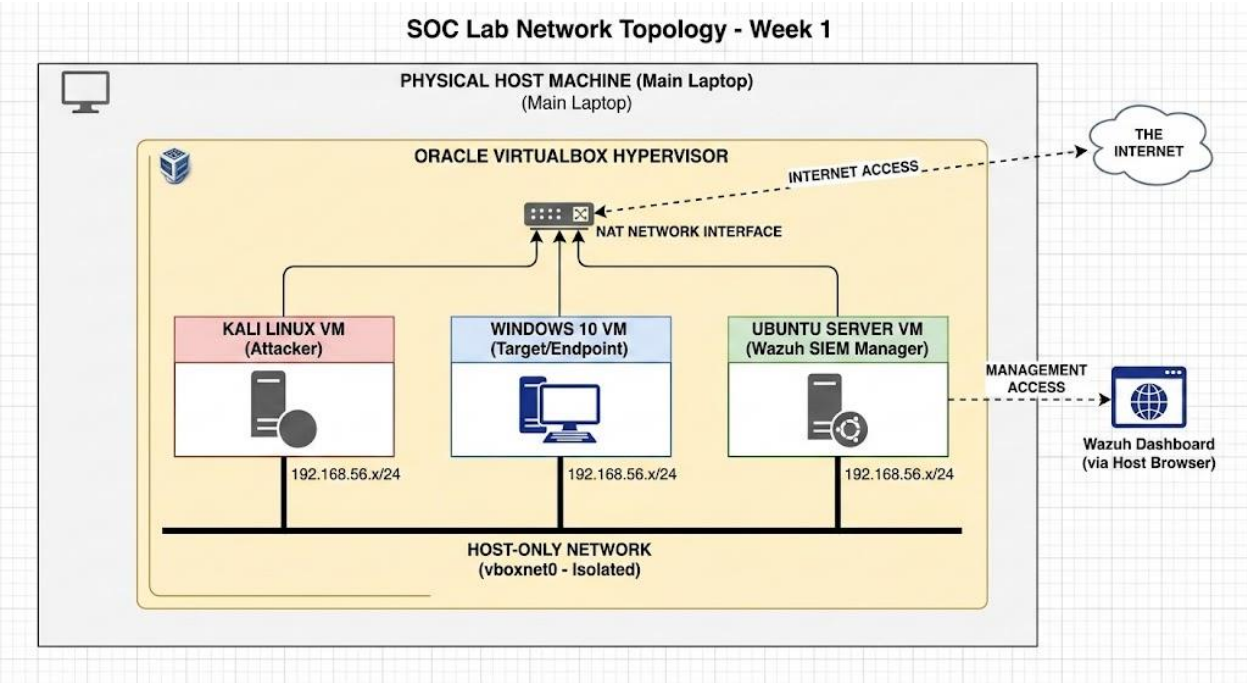
3. Lab Topology Planning

Deliverable: Planned Network Diagram

(Note: Insert your Diagram Image here)

- **Step-by-Step Explanation:**

1. **Host Machine: The main system running the hypervisor.**
2. **VirtualBox Hypervisor: The layer where my 3 VMs (Kali, Ubuntu/Wazuh, Windows 10) will reside.**
3. **Dual Adapters: Each VM is planned with two adapters:**
    - **NAT: For internet access to download updates and packages.**
    - **Host-Only: To create an isolated, secure network for the VMs to communicate without exposing the host machine to potential malware/attacks.**



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4. Practical Analysis: Wireshark Traffic Capture

Deliverable: Protocol Identification & Labeling

- **Step 1: Traffic Capture: I opened Wireshark on my host machine and captured live traffic during normal browsing.**

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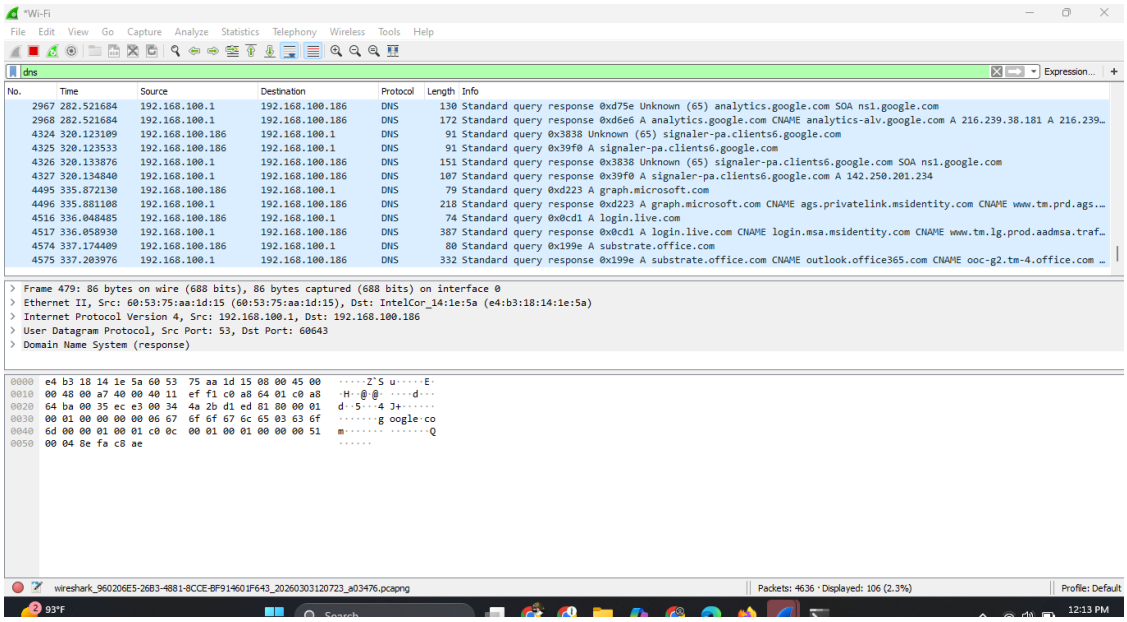
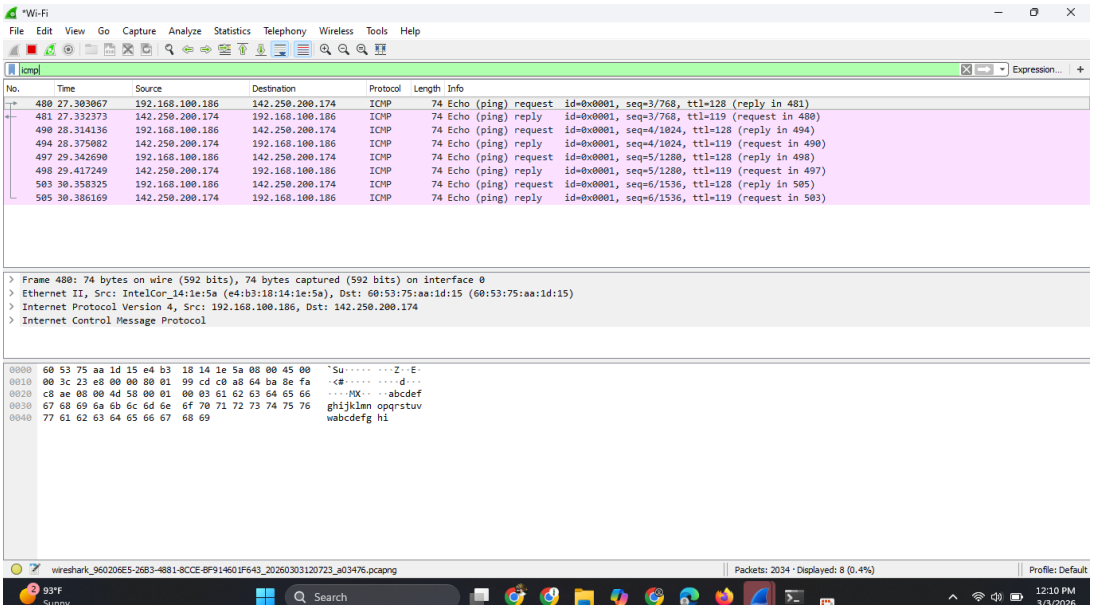


- **Step 2: Protocol Filtering: In the image above, I applied the icmp filter to isolate diagnostic traffic.**
- **Step 3: Packet Identification:**
  - **Source IP (192.168.100.186): My local machine initiating the request.**
  - **Destination IP (142.250.200.174): The remote server responding.**
  - **Info Column: Shows the Echo (ping) request and Echo (ping) reply, confirming a successful round-trip communication.**



- **Step 4: Deep Packet Inspection: The lower pane shows the Internet Control Message Protocol (ICMP) headers. This confirms the packet is at the Network Layer (Layer 3) of the OSI model.**

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5. Learning Outcomes

By the end of Day 1, I have developed the ability to:

1. **Identify whether a packet looks "Normal" or "Suspicious" based on protocol behavior.**
2. **Understand the necessity of different VirtualBox networking modes (NAT vs. Host-Only) for lab security.**
3. **Visualize how data moves from an endpoint to a management server, which is essential for the Wazuh setup starting tomorrow.**

Internship Report: Days 2 – 4: Lab Setup & SIEM Installation

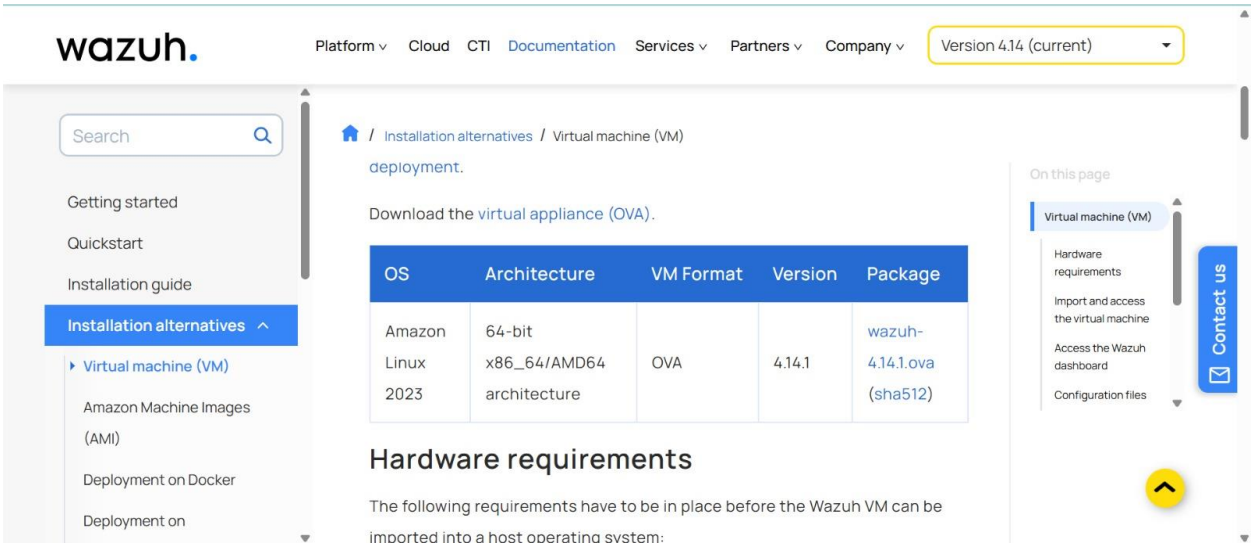
1. Day 2: VirtualBox Lab Setup

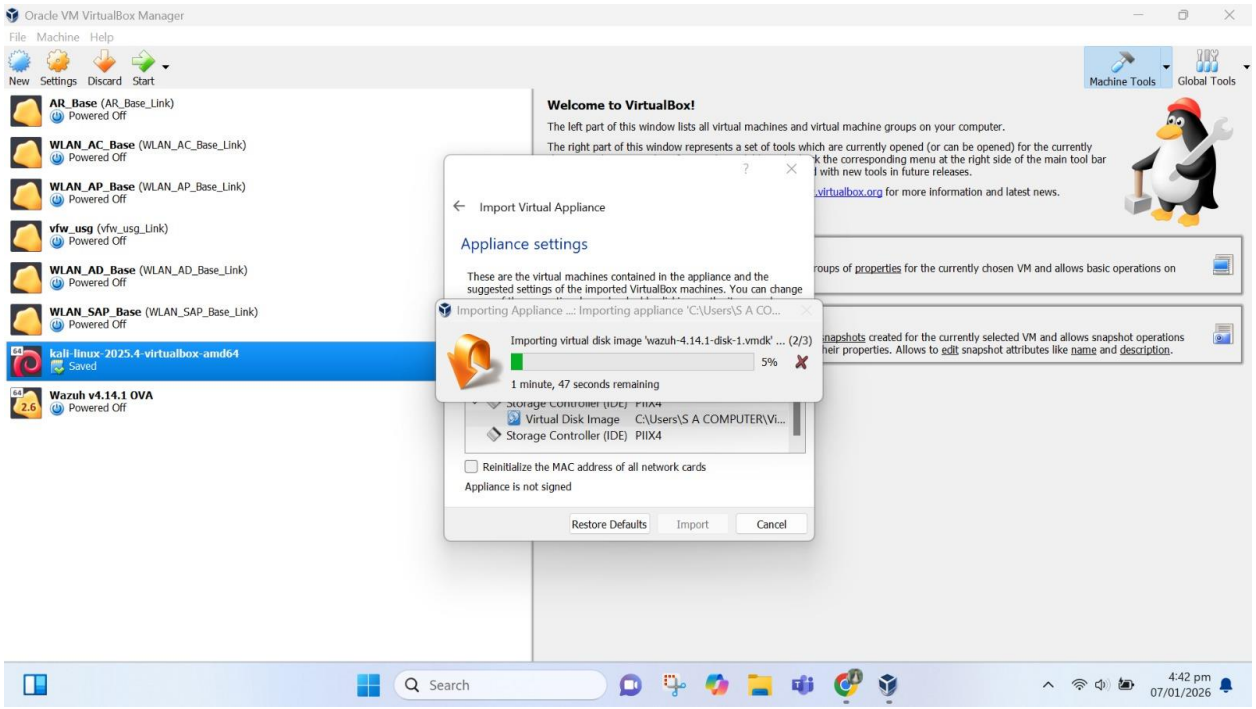
The goal was to build a simplified but realistic SOC lab consisting of three core virtual machines (VMs).

VM Specifications & Import

I created three VMs with the following allocations to ensure stability:

- **Kali Linux: Attacker machine (2 CPU cores, 4GB RAM).**
- **Ubuntu Server: Wazuh Manager (2 CPU cores, 8GB RAM preferred for memory-intensive tasks).**
- **Windows 10: Monitored endpoint (2 CPU cores, 4GB RAM).**





- **Explanation: These images show the selection and import process of the Wazuh Virtual Appliance (OVA) into VirtualBox, establishing the foundation of the lab.**

Networking Configuration

Networking is critical for lab functionality. Each machine was configured with two adapters:

1. **Host-Only Adapter: For internal communication between VMs and the host machine.**

2. NAT Adapter: To allow the machines to reach the internet for necessary downloads and updates.

Connectivity & Snapshots

- IP Verification: Confirmed that VMs received IPs on both the Host-Only subnet (e.g., 192.168.56.x) and the NAT subnet.
- Snapshots: Once connectivity was verified, I took snapshots of each VM as a "safety net" before proceeding with the Wazuh installation.

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2. Days 3-4: Wazuh Manager Installation

Wazuh serves as the SIEM (Security Information and Event Management) platform, collecting log data, running detection rules, and surfacing events for investigation.

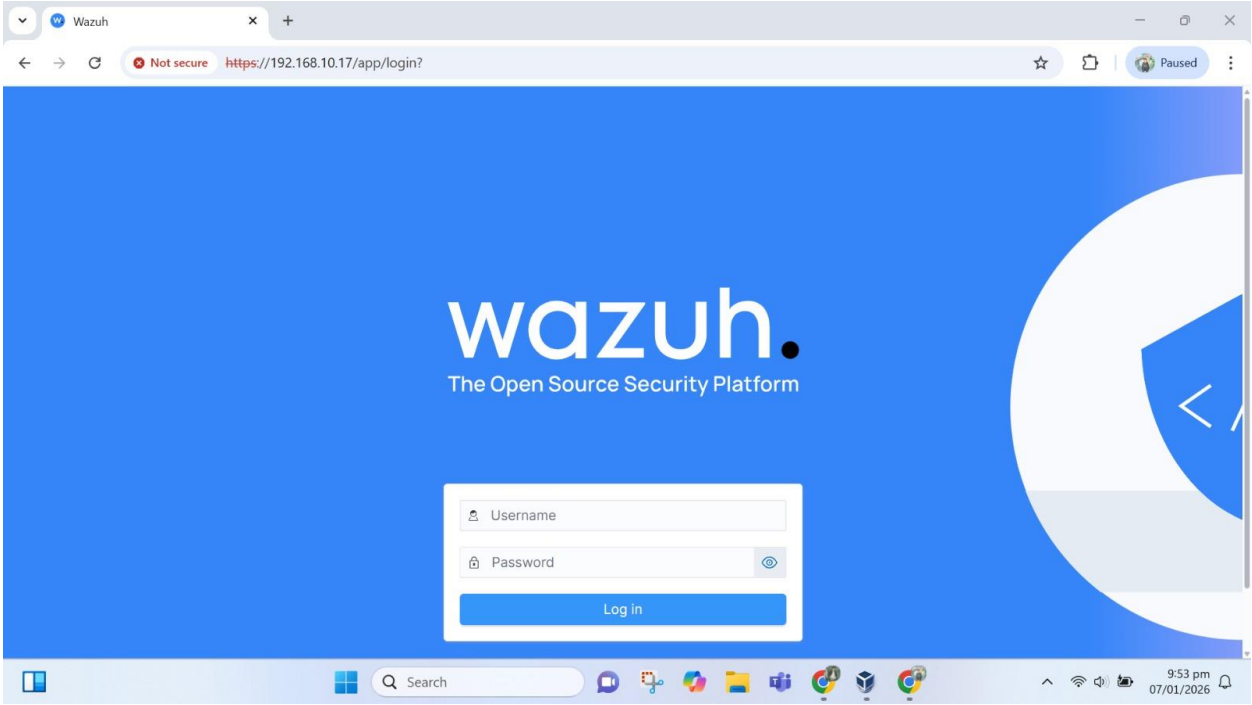
The Wazuh Stack

The installation on the Ubuntu Server includes three vital components:

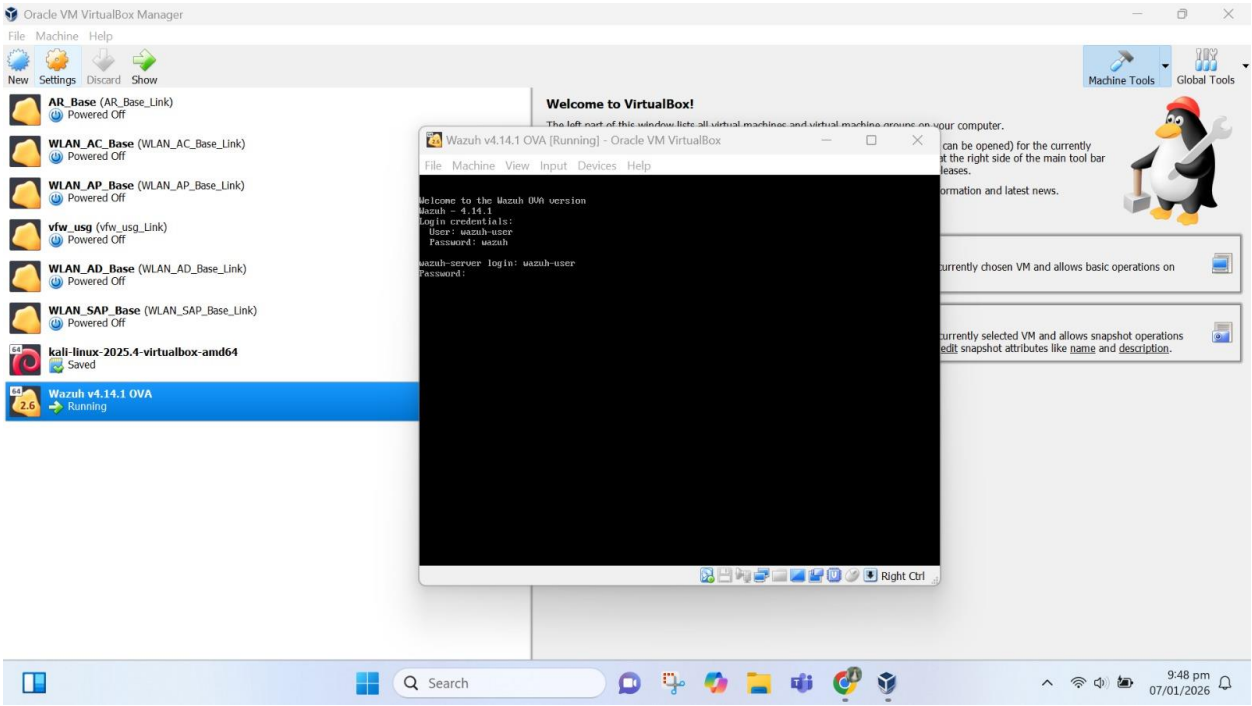
1. Wazuh Manager: Processes logs and generates alerts.
2. Wazuh Indexer: Stores log data in a searchable format.
3. Wazuh Dashboard: The web interface for analysts.

Installation Process

- Static IP: Assigned a static IP to the Ubuntu VM on the Host-Only network to prevent agents from losing connection if the IP changes.
- Quickstart Script: Followed the official Wazuh documentation using the quickstart installation script to configure all three components automatically.



- **Explanation: This screenshot shows the Wazuh Server console after booting, providing the default credentials and access URL.**



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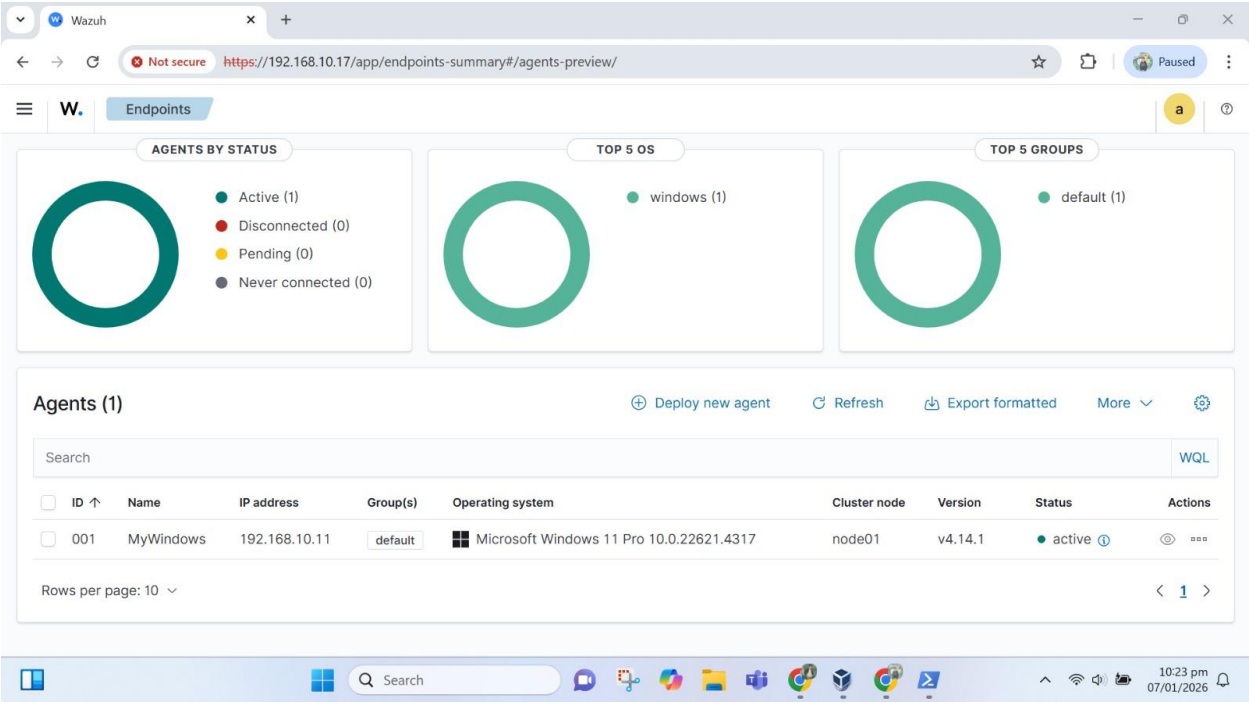
Service Verification & Dashboard Access

- **Service Check: Verified that all services (wazuh-manager, wazuh-indexer, wazuh-dashboard) were active and running using the systemctl status command.**
- **Web Access: Logged into the dashboard via the browser using the Host-Only IP.**



- **Explanation: These images show the HTTPS login page and the successful loading of the Wazuh Dashboard Overview.**

Internship Report: Days 5 – 6: Wazuh Agent Deployment

1. Objective

The objective of these two days was to connect the Kali Linux and Windows 10 endpoints to the Wazuh Manager. This establishes a client-server model where agents collect log data and system events to send back to the central manager for analysis.

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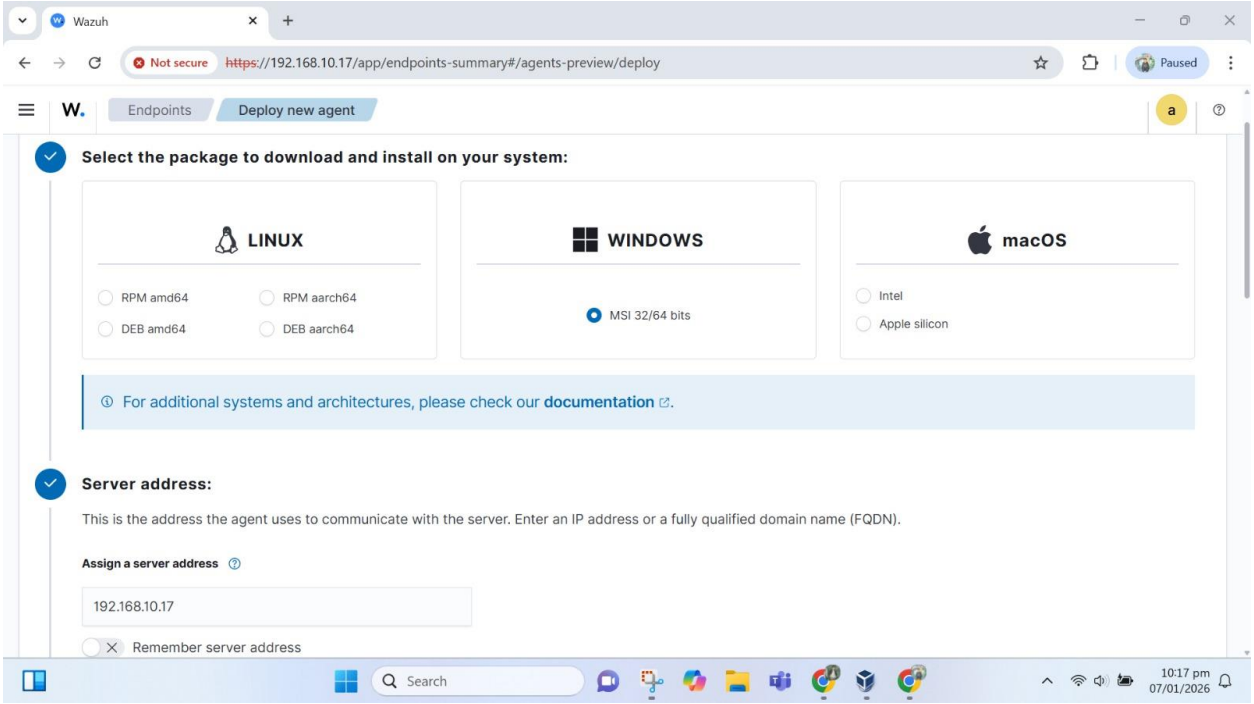
2. Agent Deployment Process

Using the Deployment Wizard

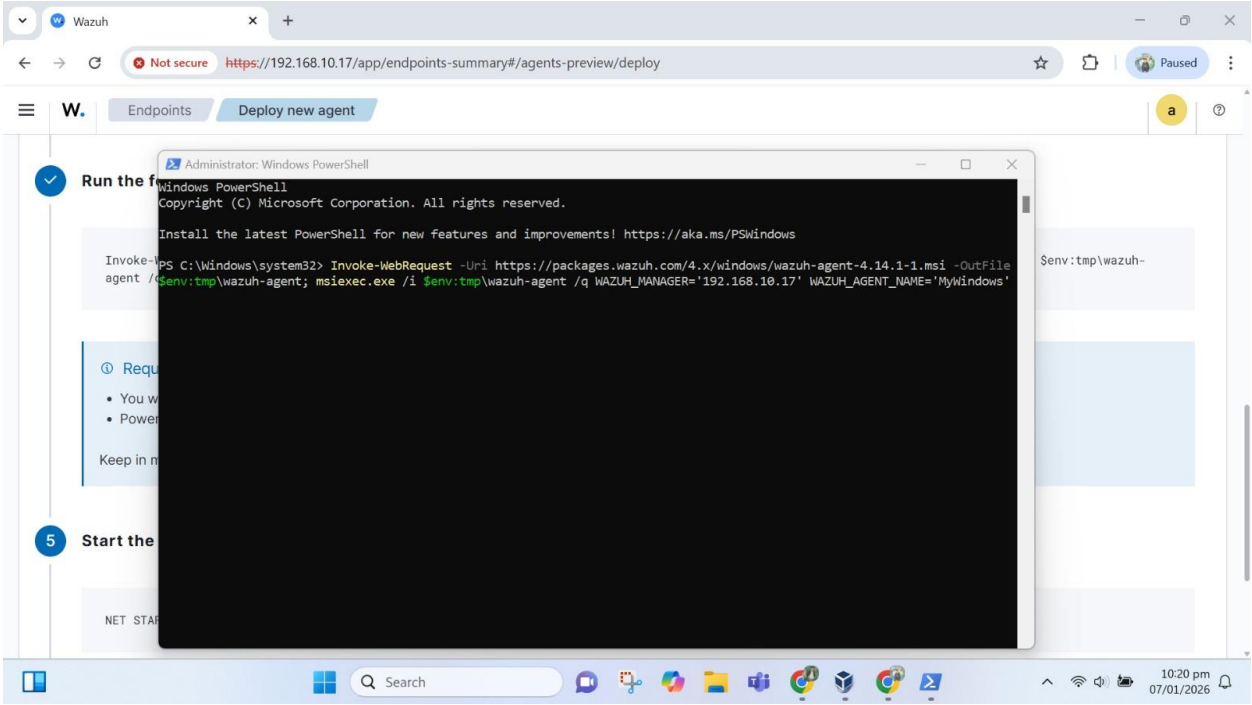
To ensure the correct configuration, I used the Wazuh dashboard's built-in deployment wizard. This tool pre-fills the Manager's static IP and selects the appropriate package for the target Operating System.



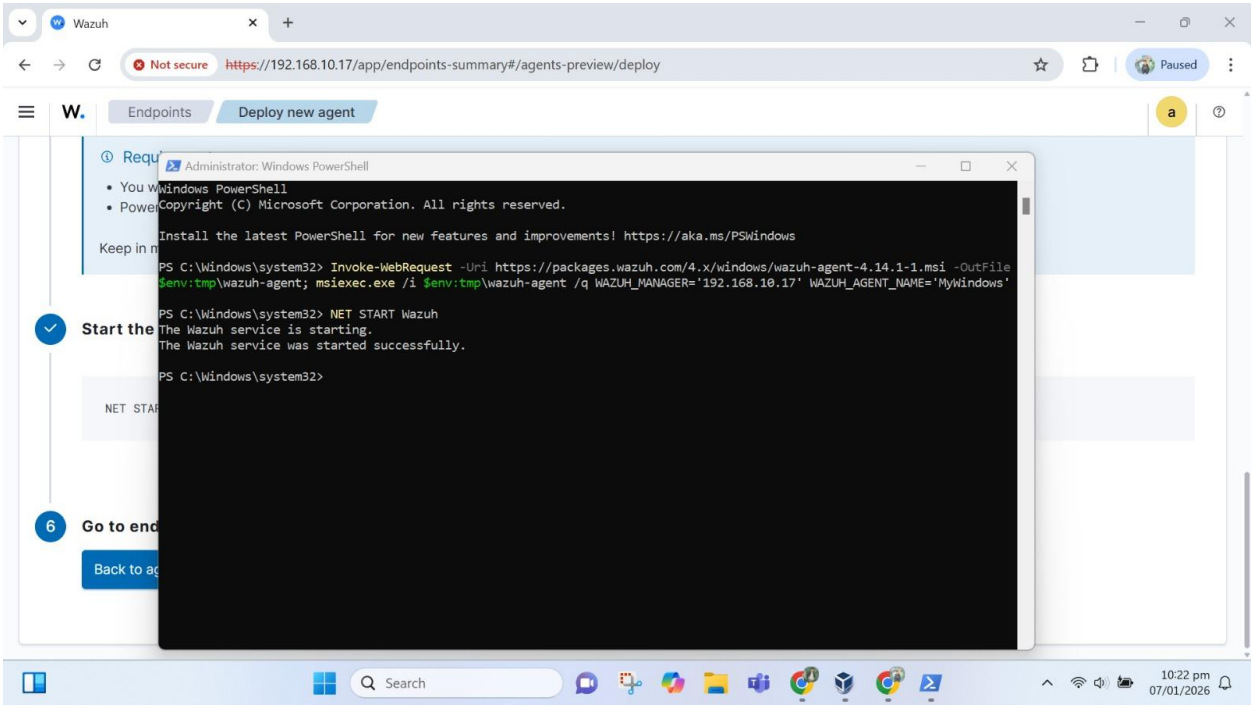
- **Explanation: These screenshots show the Wazuh "Deploy new agent" wizard where I selected the Windows MSI and specified the Manager IP (192.168.10.17).**

Windows 10 Installation

- **Execution: I ran the generated PowerShell command in an elevated (Administrator) session on the Windows 11 VM.**
- **Service Activation: Started the Wazuh service to begin the telemetry stream.**



- **Explanation: The first image shows the PowerShell installation in progress. The second shows the local Wazuh Agent interface on the Windows machine confirming the status is "Running."**



Kali Linux Installation

- **Execution: Following the same wizard process, I deployed the agent on the Kali Linux VM.**
- **Verification: Confirmed the agent appeared as Active in the dashboard.**

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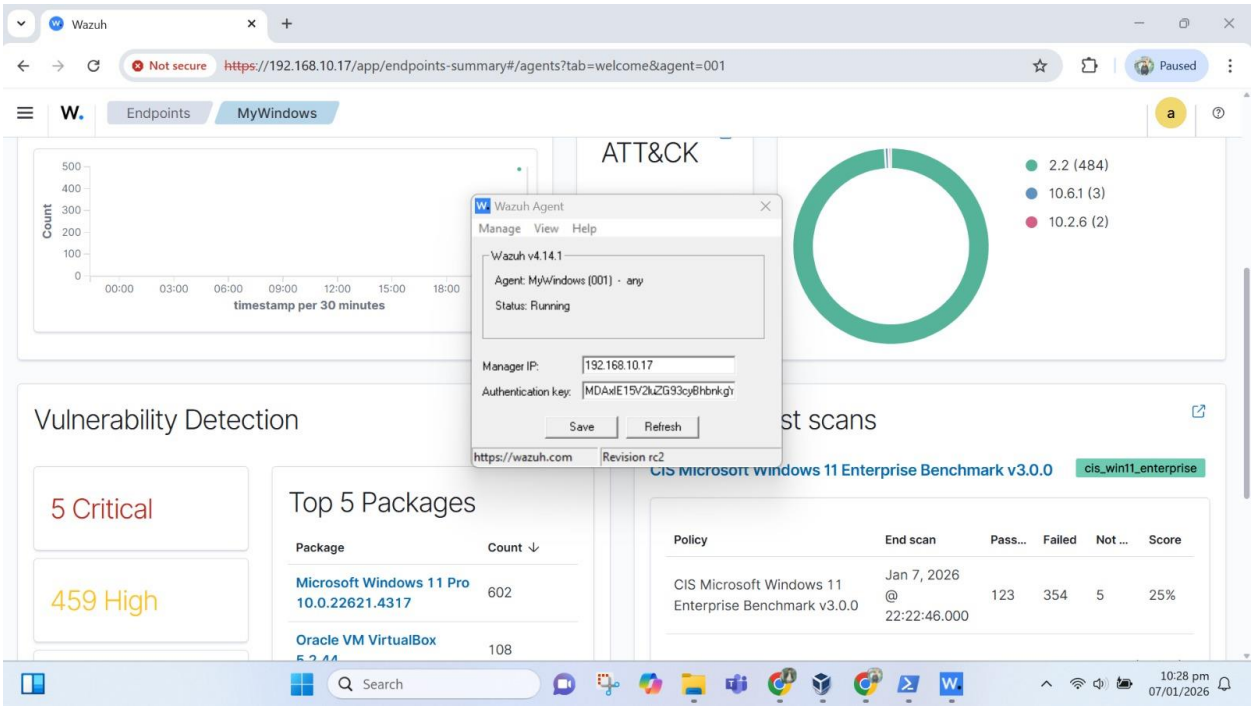
3. Configuration and Connectivity

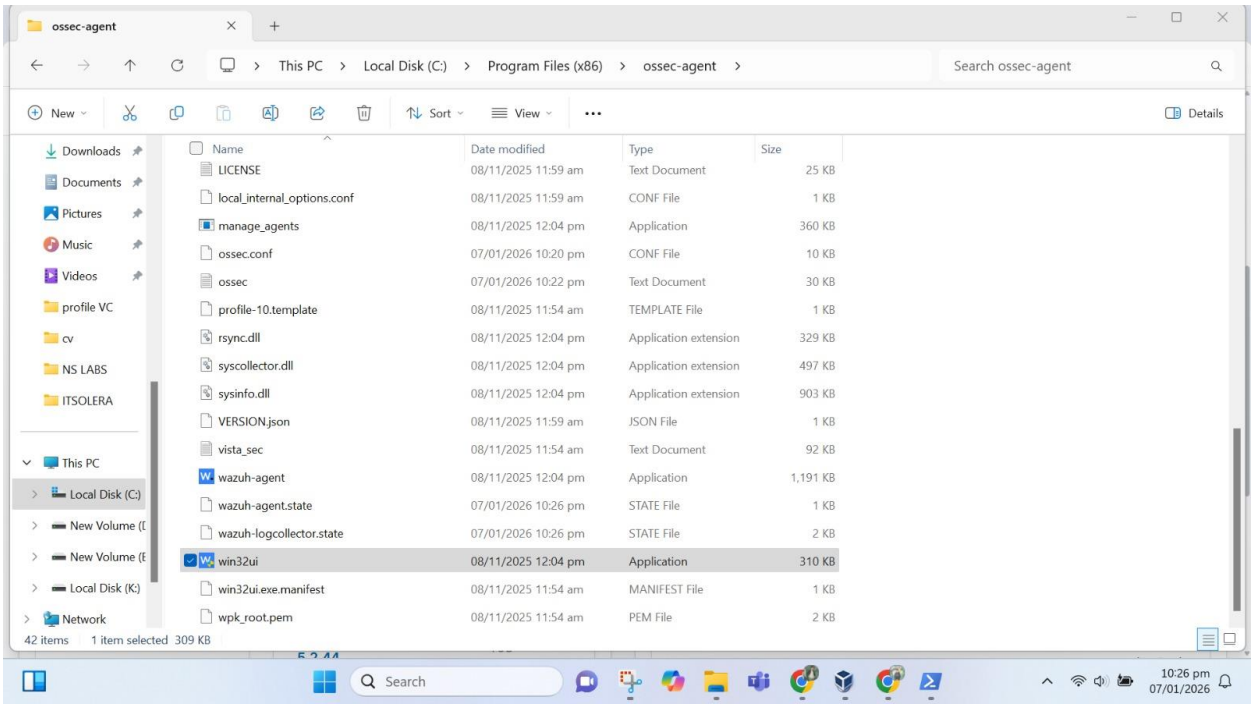
The ossec.conf File

I reviewed the ossec.conf file, which is the primary configuration file for the agent. It defines:

- **Server Block: Where the log data is sent (Manager IP).**
- **Syscheck Block: Controls File Integrity Monitoring (FIM).**
- **Localfile Blocks: Specifies which OS log files the agent should monitor.**

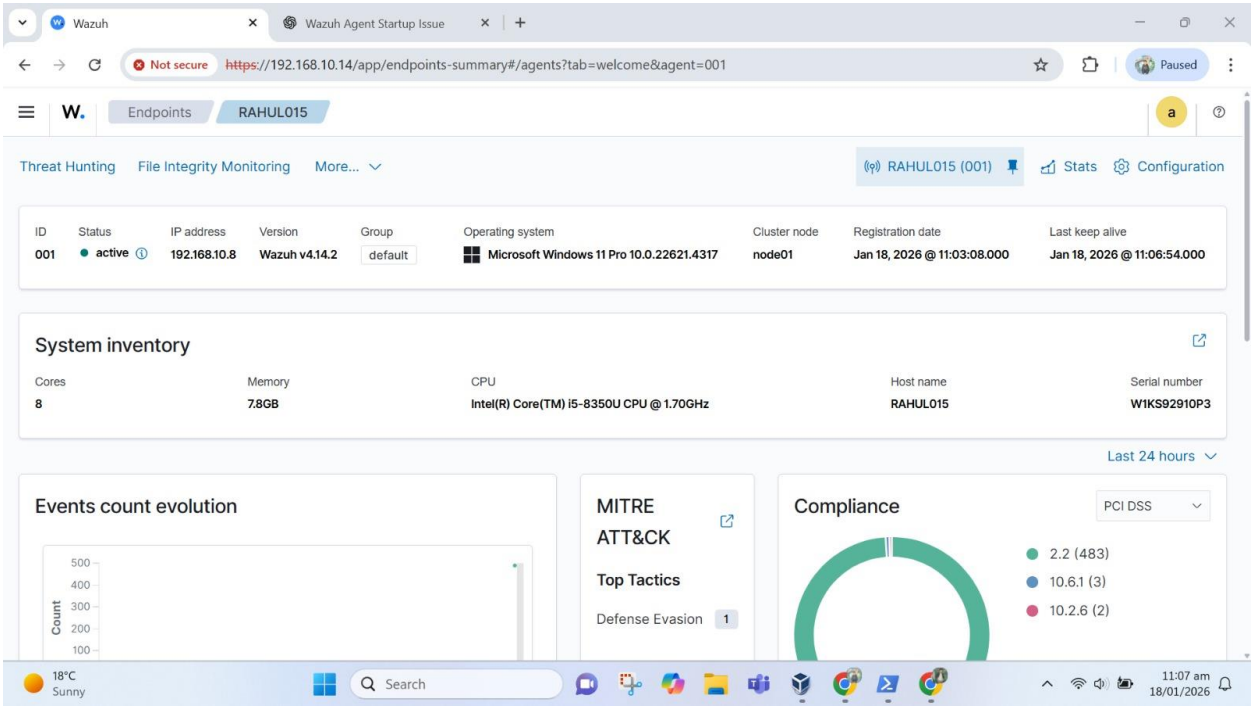


- **Explanation: This screenshot shows the ossec.conf file where the monitoring parameters are defined for the endpoint.**



Active Status Confirmation

Both agents successfully registered and appeared as Active simultaneously in the dashboard.



- **Explanation: This dashboard view confirms that the Windows agent is connected and "Active," proving the communication pipeline is open.**

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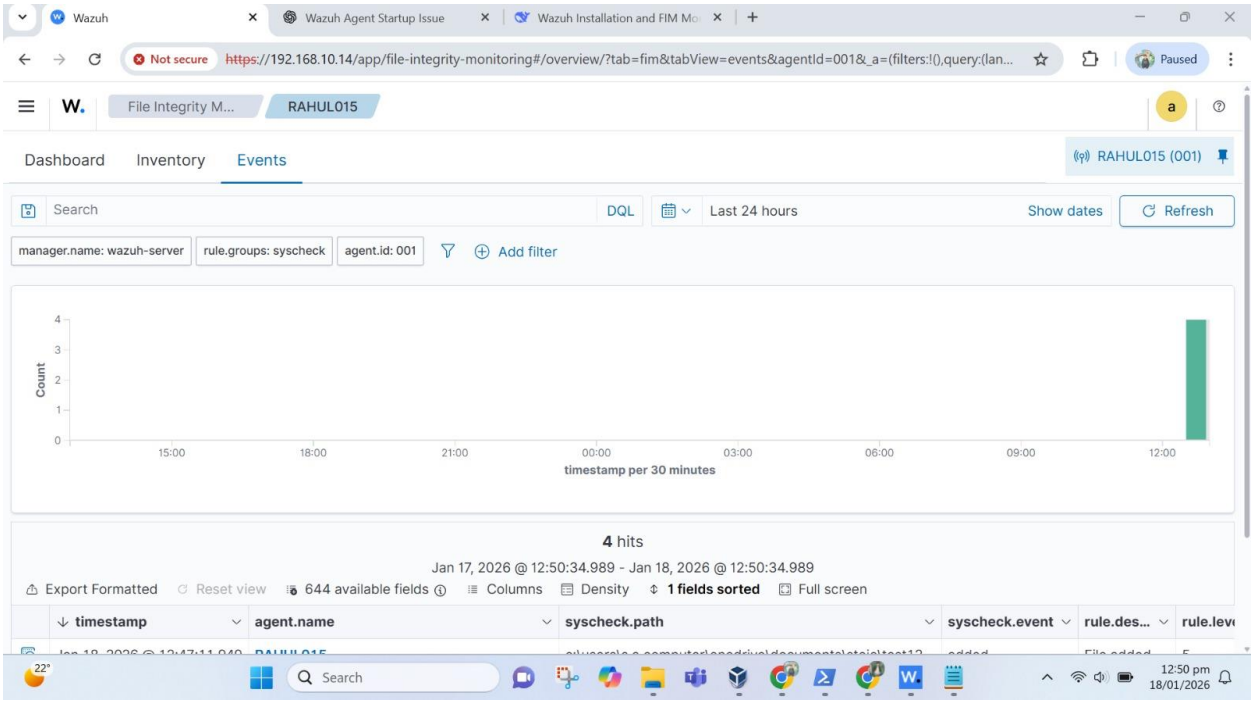
4. Generating Test Alerts

To verify the full pipeline (Endpoint -> Manager -> Dashboard), I generated test alerts.

- **Method: Attempted several failed SSH logins to the Ubuntu Server from the Kali Linux VM using incorrect passwords.**
- **Result: After a short delay, the failed login attempts triggered security alerts visible in the Wazuh "Security Events" tab.**



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5. Learning Outcomes

- **Client-Server Model: Understood that agents initiate the connection to the manager to send data; the manager does not reach out to them.**
- **Visibility: Gained an appreciation for how centralized monitoring across multiple machines makes unusual behavior easier to spot.**
- **Troubleshooting: Learned that firewall ports (1514 and 1515) must be open on the manager for agents to communicate.**

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Internship Report: Day 7 – Dashboard Navigation & Alert Triage

1. Objective

The goal of Day 7 was to shift from "Setup Mode" to "Analysis Mode." By navigating the Wazuh dashboard, practicing query writing, and mapping events to the MITRE ATT&CK framework, I began building a baseline of "normal" activity for this environment.

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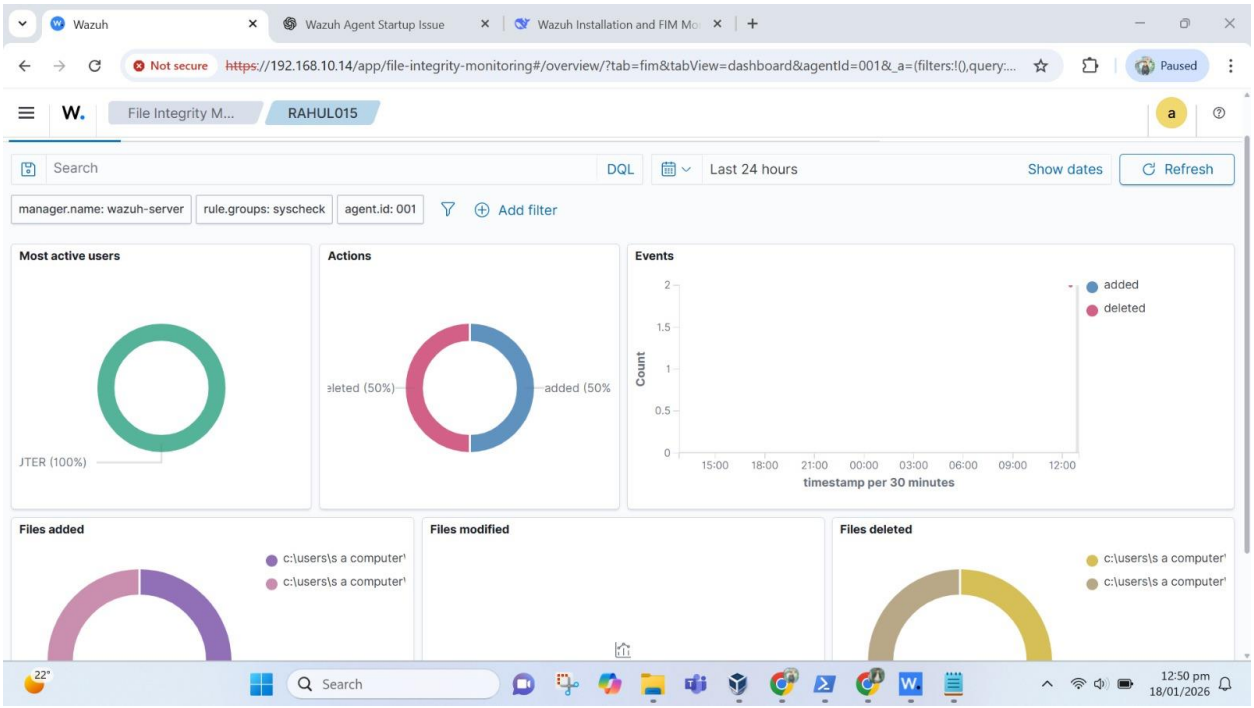
2. Navigating the Security Dashboard

Security Alerts & Rule Levels

I spent time analyzing individual alerts to understand the Wazuh severity hierarchy. Wazuh rules range from 0 to 15:

- **0–3: Informational.**
- **4–7: Low severity (e.g., routine system changes).**
- **8–11: Medium severity.**
- **12–15: High/Critical (e.g., potential attacks).**



- **Explanation: This screenshot shows the Security Events log. Each entry displays the Rule Level, the Agent involved, and the specific timestamp of the event.**

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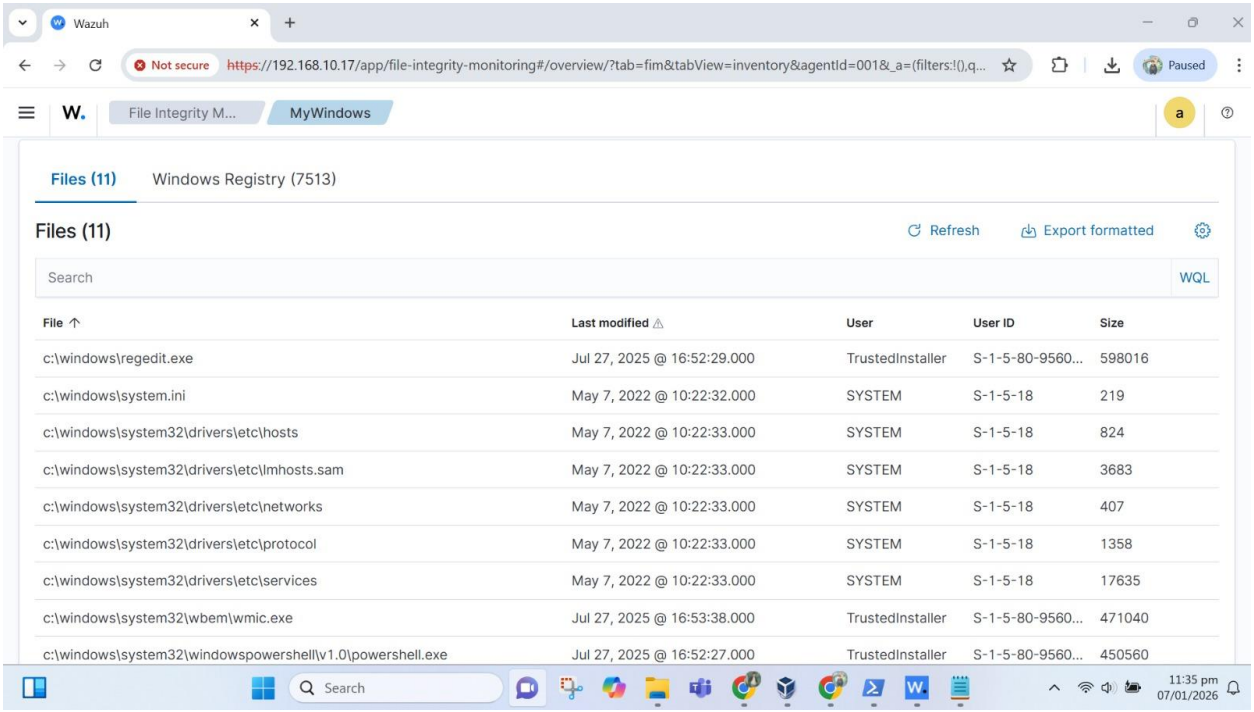
3. Advanced Analysis Modules

Threat Hunting & Query Writing

To avoid "alert fatigue," I practiced using the Threat Hunting module to filter data. I learned to filter by:

- **Specific Agent names.**
- **Rule levels (e.g., only Level 7 and above).**
- **Specific time ranges to pinpoint when a "spike" in activity occurred.**



- **Explanation: The File Integrity Monitoring (FIM) dashboard. This visualizes file additions and deletions, allowing an analyst to see trends at a glance rather than scrolling through thousands of lines of text.**

MITRE ATT&CK Mapping

I reviewed the MITRE ATT&CK module, which categorizes alerts into known attacker techniques. Mapping alerts to this framework is vital for understanding the *intent* of an attacker (e.g., Persistence vs. Privilege Escalation).

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4. Deliverable: Alert Triage Findings

Based on the laboratory environment, I identified the top alert types. Below is the triage summary:

Rule ID	Level	Description	Analyst Assessment
550	7	Integrity checksum changed (File modified).	Routine: Likely a system log update or manual config change.
554	5	File added to the system.	Normal: Triggered during my manual test file creation (test1.txt).
594	7	File deleted from the system.	Normal: Created during cleanup of test files.
5710	5	Attempt to login using a non-existent user.	Suspicious: Triggered by my failed SSH tests from Kali Linux.
5712	10	SSHD brute force attempt.	Concern: High frequency of failed logins indicates a potential attack.

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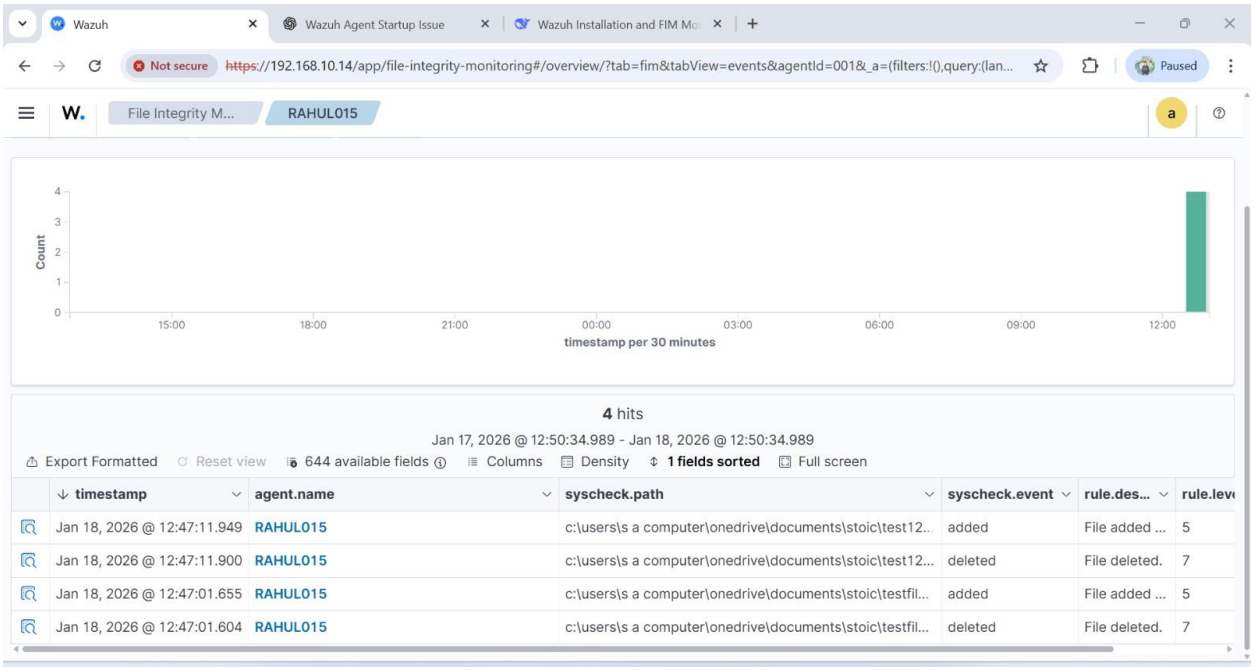

- **Explanation: This triage screen shows the Rule Level 7 (Deleted) and Rule Level 5 (Added) alerts. By clicking the dropdown arrow, I can see the full JSON metadata for the event.**